Practical 24

Exercise

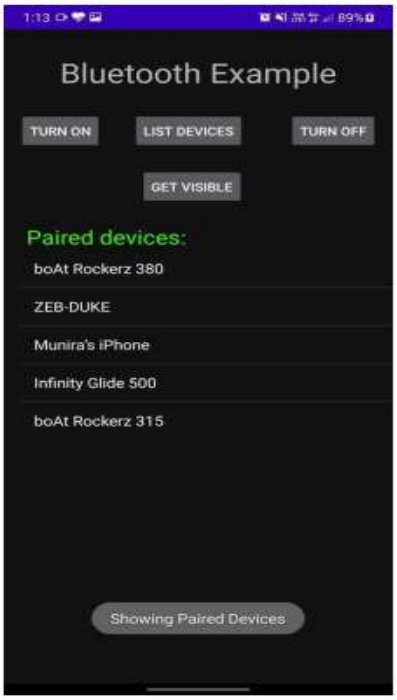
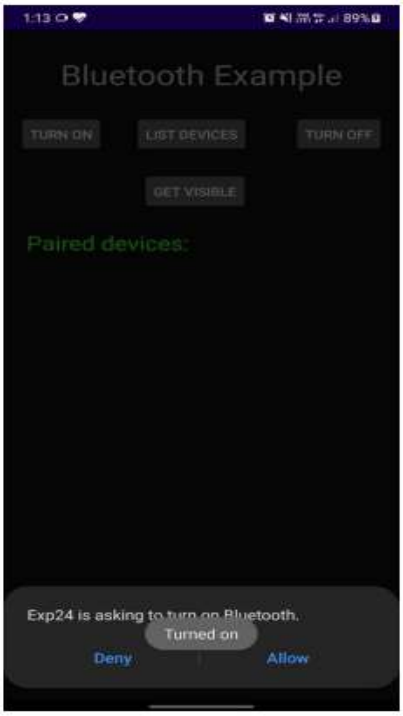
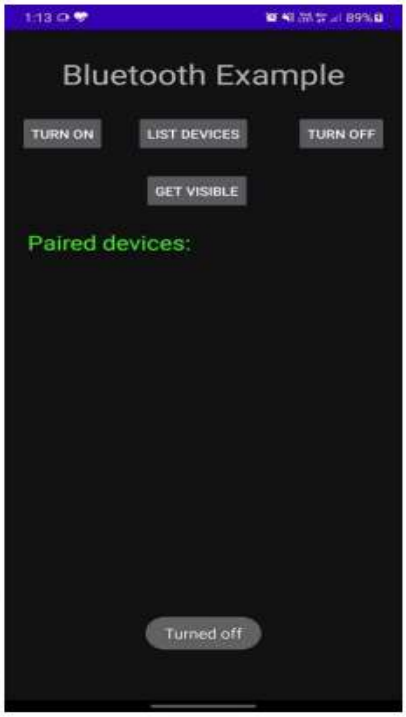
1.Activitymain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/turn_on_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Turn On Bluetooth"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp"/>

    <Button
        android:id="@+id/get_visible_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Make Visible"
        android:layout_below="@id/turn_on_button"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/list_devices_button"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="List Devices"
        android:layout_below="@id/get_visible_button"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/turn_off_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Turn Off Bluetooth"
        android:layout_below="@id/list_devices_button"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"/>

</RelativeLayout>

import android.bluetooth.BluetoothAdapter;
import android.bluetooth.BluetoothDevice;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Set;
```

```
public class MainActivity extends AppCompatActivity {

    private BluetoothAdapter bluetoothAdapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        bluetoothAdapter = BluetoothAdapter.getDefaultAdapter();

        Button turnOnButton = findViewById(R.id.turn_on_button);
        Button getVisibleButton = findViewById(R.id.get_visible_button);
        Button listDevicesButton = findViewById(R.id.list_devices_button);
        Button turnOffButton = findViewById(R.id.turn_off_button);

        turnOnButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                turnOnBluetooth();
            }
        });

        getVisibleButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                makeVisible();
            }
```

```
        });

        listDevicesButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                listDevices();
            }
        });

        turnOffButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                turnOffBluetooth();
            }
        });
    }

    private void turnOnBluetooth() {
        if (bluetoothAdapter != null) {
            if (!bluetoothAdapter.isEnabled()) {
                Intent enableBtIntent = new Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
                startActivityForResult(enableBtIntent, 0);
                Toast.makeText(getApplicationContext(), "Bluetooth turned on",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getApplicationContext(), "Bluetooth is already on",
Toast.LENGTH_SHORT).show();
            }
        } else {
```

```
            Toast.makeText(getApplicationContext(), "Bluetooth not supported on this device",
Toast.LENGTH_SHORT).show();
        }
    }

    private void makeVisible() {
        if (bluetoothAdapter != null) {
            Intent discoverableIntent = new Intent(BluetoothAdapter.ACTION_REQUEST_DISCOVERABLE);
            discoverableIntent.putExtra(BluetoothAdapter.EXTRA_DISCOVERABLE_DURATION, 300);
            startActivity(discoverableIntent);
        }
    }

    private void listDevices() {
        if (bluetoothAdapter != null) {
            Set<BluetoothDevice> pairedDevices = bluetoothAdapter.getBondedDevices();
            if (pairedDevices.size() > 0) {
                StringBuilder devicesList = new StringBuilder();
                for (BluetoothDevice device : pairedDevices) {
                    devicesList.append(device.getName()).append("\n");
                }
                Toast.makeText(getApplicationContext(), "Paired devices:\n" + devicesList.toString(),
Toast.LENGTH_LONG).show();
            } else {
                Toast.makeText(getApplicationContext(), "No paired devices found",
Toast.LENGTH_SHORT).show();
            }
        }
    }
```

```
    private void turnOffBluetooth() {
        if (bluetoothAdapter != null) {
            if (bluetoothAdapter.isEnabled()) {
                bluetoothAdapter.disable();
                Toast.makeText(getApplicationContext(), "Bluetooth turned off",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getApplicationContext(), "Bluetooth is already off",
Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```